

Hotel Booking Cancellation Prediction

A Business Intelligence Solution for Revenue Management

Final Project Presentation

Group 4: Asaad Razeq | Said Huner | Nora Yong | Cassidy Taillon

BIA 5450 Capstone | Humber Polytechnic | Professor Fayruz Kibria

The Business Problem

Why a fixed overbooking percentage does not solve cancellations

Between 15 and 40 percent of hotel bookings cancel before arrival

- A room cancelled close to arrival often cannot be resold at any price

The industry answer is a static overbooking percentage

- It treats every reservation as carrying the same risk
- It leaves money on the table when the book is firm, and walks guests when it is not

The signals are already in the reservation system

- Lead time, channel, deposit type, country and rate all move cancellation risk
- A single percentage cannot tell a 200 day online booking from a 5 day direct one

What is missing

- A per booking probability, and that probability expressed in dollars a manager acts on

Who uses the output

Revenue Manager, works the ranked list daily

General Manager, watches the exposure figure weekly

Marketing Manager, compares channels after cancellations

Front Desk Manager, prepares for likely no shows

Director of Sales, opens deposit conversations on groups

Scope Statement

What we committed to build, and what we deliberately left out

In scope	Out of scope
Binary cancellation prediction on the Hotel Booking Demand dataset	Multi class no show or late arrival prediction
Two Portuguese hotels, one city and one resort, 2015 to 2017	Hotels outside Portugal, or data after 2017
Logistic regression baseline plus two tree based models	Deep learning and neural network approaches
Power BI dashboard for the revenue manager	Mobile application or real time push notification
Manually triggered pipeline run and dashboard refresh	Live retraining on streaming booking data
Recommended actions by risk band	Automated execution of those actions in the reservation system

Constraints: the data is historical and covers two properties in one country, so the percentages are properties of this dataset and not current market rates. No live database was available, so the analytical layer hands off to Power BI through a schema controlled file.

Analytics Questions

All four types, each tied to a decision it enables

Type	Question	Decision it enables
Descriptive	What is our cancellation rate by quarter, hotel and market segment?	Shows where the loss concentrates
Diagnostic	Why do online travel agent bookings cancel far more often than direct bookings?	Points to deposit policy as the lever
Predictive	Given a new booking, what is the probability it will be cancelled?	Produces the per booking risk score
Predictive	How many bookings on the books are likely to cancel, and what is the exposure?	Sizes the forward risk for overbooking
Prescriptive	Which bookings get a retention offer, and which get a deposit request?	Generates the daily action list
Prescriptive	What overbooking level should we set for a given arrival date?	Converts the forecast into inventory

The dashboard answers the descriptive and diagnostic questions through its slicers, and the action list page answers the prescriptive ones.

Data Sources and Key Entities

One source of record, five business entities, one new entity

Hotel Booking Demand dataset

- 119,390 bookings, 32 fields, 16.07 MB
- One city hotel and one resort hotel in Portugal
- Arrivals from July 2015 to August 2017

Why we judged it credible

- Published as a peer reviewed data article, Antonio and others 2019
- Clearly defined cancellation target plus booking, guest, stay and channel fields
- Long enough to support a split on time rather than at random

Where it is weak

- Historical, and concentrated in two properties in one country

Entity	System of record in production
Booking	Property management system
Hotel and stay	Property management system
Guest	Property management and CRM
Distribution	Central reservation system and OTA
Outcome	Property management system
Scored booking	Created by our analytical layer

Process and Data Flow

Every step with the data it consumes and the data it produces



Steps 1 to 10 run in sequence: each step waits for the data the previous step produces. Step 11 is conditional, the action depends on the risk band.

Data Manipulation and Output

Documented rules, nothing deleted silently

119,390

Source records

118,563

Retained, 99.31 percent

827

Separated into the audit file

0

Missing cells after cleaning

Issue found in the audit	Decision	Reason
company missing in 94 percent of rows	Keep a presence flag, drop the identifier	Presence is meaningful, the raw code is not usable
31,994 duplicate looking rows	Retain and flag	No booking identifier exists to prove a repeat is an error
Zero guests, zero nights, invalid rates	Reject into an audit file with a reason	827 unique rows, preserved rather than deleted
Reservation status reveals the outcome	Exclude from the model ready dataset	Otherwise the model looks perfect and is useless

Three outputs: an analytics dataset of 48 fields, a model ready dataset of 37 booking time fields, and a Power BI base table under schema version 1.1. A reproducibility test regenerated all of them from the untouched source and matched every row count and value.

Solution Design and Methodology

Python does the modelling, Power BI does the deciding

Binary classification on the cancellation flag

- Booking time fields only, so a new booking can be scored the moment it arrives

Split on time, not at random

- Train on 94,850 earlier bookings, test on 23,713 later ones
- A random split would let the model learn from August and be tested on June

Three algorithms compared, not one chosen up front

- Logistic regression as the explainable baseline, random forest and gradient boosting
- Primary metric is ROC-AUC, because the job is to rank bookings by risk

The output is money, not probability

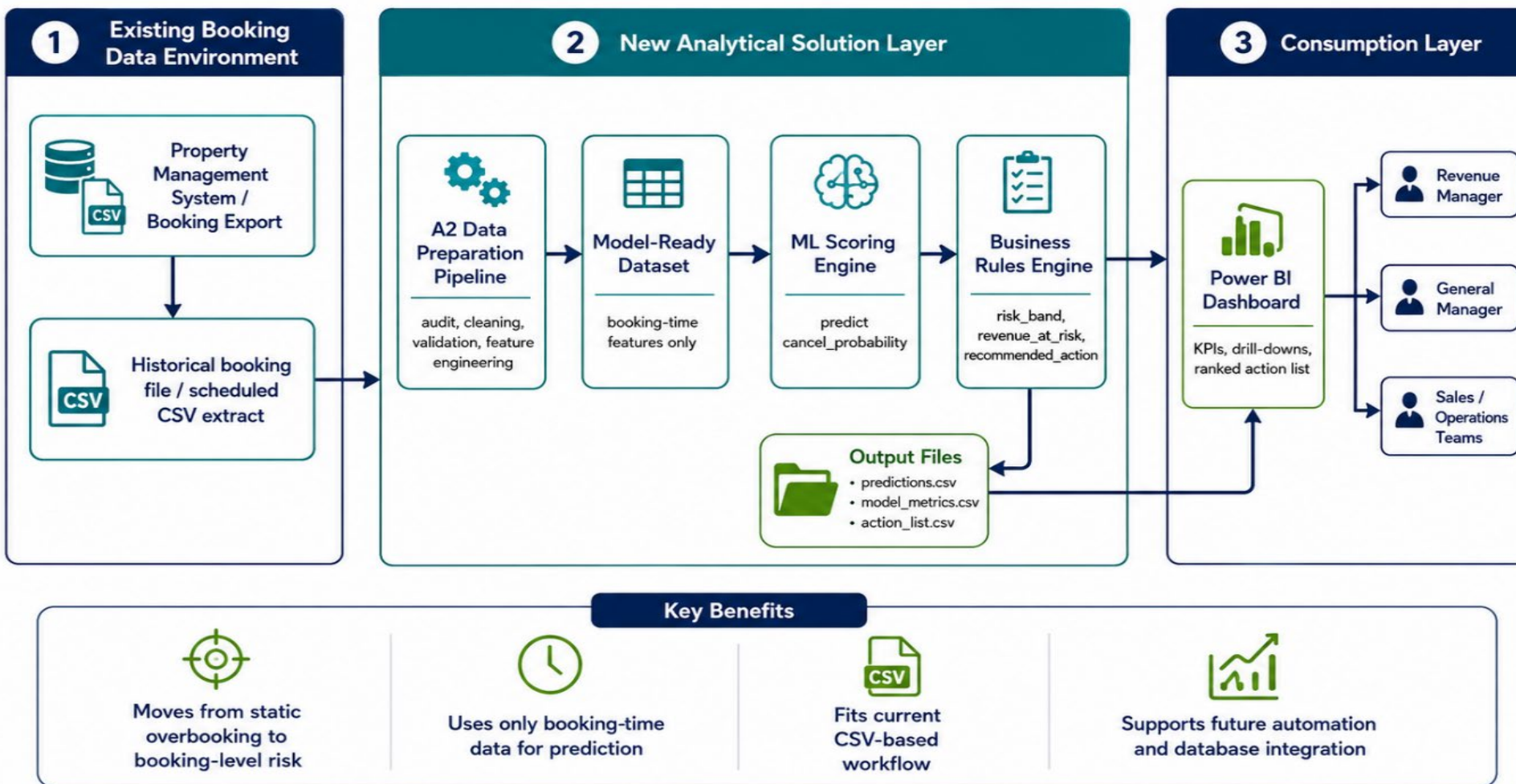
- Revenue at Risk = booking value multiplied by cancellation probability

Decision	Alternative we rejected
Temporal split	Random split, gives an optimistic score
ROC-AUC as primary	Accuracy, misleading at a 41 percent rate
Three models compared	One model picked up front
Revenue at Risk in dollars	Showing the raw probability
Power BI on one table	Charts in Python a user cannot filter

Fit into the Existing IT Architecture

A new analytical layer, nothing existing is replaced

Figure 3. Proposed Solution Fit within the Existing IT Architecture



Implementation: Three Models Compared

Same features, same split, same test period

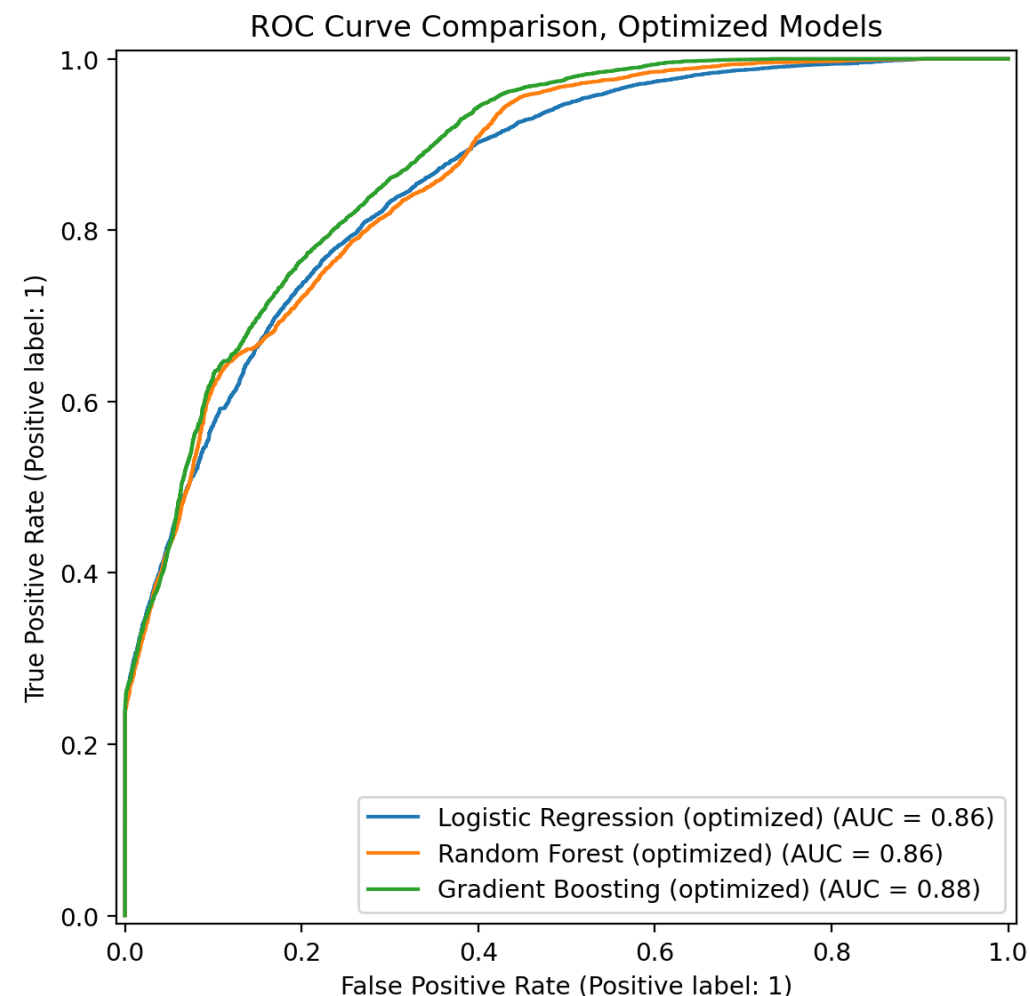
Model	ROC-AUC	Precision	Recall	Time
Gradient Boosting	0.8798	0.819	0.617	5 s
Random Forest	0.8644	0.774	0.658	6 s
Logistic Regression	0.8598	0.710	0.753	4 s

Final model: Gradient Boosting

- ROC-AUC 0.8798 against a target of 0.80

All three sit within 0.02 of each other

- The choice of features mattered far more than the choice of algorithm
- The whole pipeline trains in about five seconds



Outcome Testing

Measured against the targets agreed at the scope stage

Requirement	Target	Achieved	Met
Ranks cancellation risk reliably	ROC-AUC at or above 0.80	0.8798	Yes
Catches most real cancellations	Recall at or above 0.70	0.8226	Yes
Flagged bookings are usually correct	Precision at or above 0.65	0.6857	Yes
Risk drivers can be explained	Per feature driver reporting	Yes, permutation importance	Yes
Refresh fits a working session	End to end under 30 minutes	5 seconds to train the final model	Yes
Drill down by hotel, segment, channel, country and season	Available in the dashboard	Built and verified	Yes
Dashboard reachable by URL through the Power BI Service	Published report	Built in Desktop, not published	No

Three business scenarios were traced end to end: a short lead direct booking scored 0.002 and is left alone, a long lead summer booking scored 0.991 and triggers a deposit request

Six of seven targets met. The one open item is publication to the Power BI Service, and we report it as open rather than as complete

From Probability to Money

What the revenue manager actually receives

23,713

Bookings scored in the test period

11,656

On the action list

\$4.28 M

Total Revenue at Risk

\$3.65 M

Revenue at Risk on the action list

Risk band	Recommended action	Bookings	Revenue at Risk	Really cancelled
High	Retention Offer	4,017	\$1,629,574	0.724
Very High	Request Deposit	3,306	\$1,316,173	0.934
Watch	Watch List	4,333	\$700,561	0.460
Low	Monitor	12,057	\$629,650	0.143

The bands separate cleanly

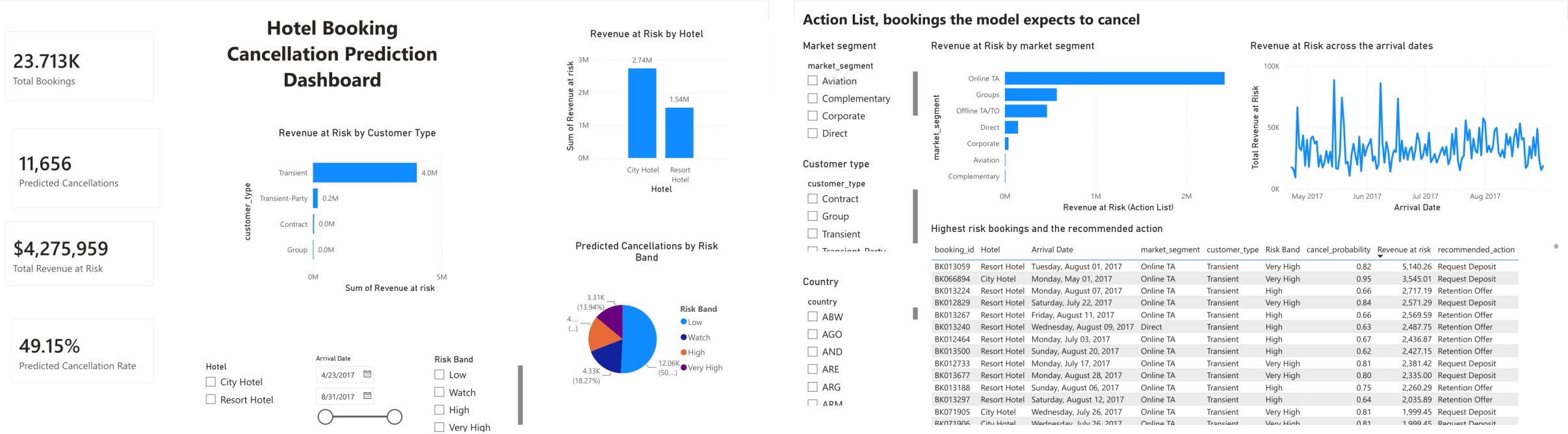
Very High cancels at 0.934, Low at 0.143, more than six times apart

A manager working only the Very High band is working a queue where nine in ten will cancel

Total booking value in the period was \$12.10 M

The Dashboard

Where the revenue manager works



Testing the Reporting Layer

The most useful failure we found

We tested the dashboard the way we tested the model

- Nine control totals computed from the scoring output, then recomputed from the dashboard

The first build failed three of them

- It loaded 9,739 bookings where the final model flags 11,656, so it was reading a stale file
- The headline card ignored every slicer, because filter clearing had been added
- It reported about 8 percent cancellations against an actual 40.97%

One root cause, one fix

- The report had been built on two tables covering two different populations
- Rebuilt on the single scored booking table. All nine measures now reconcile

Control total	Required value
Total Bookings	23,713
Predicted Cancellations	11,656
Actual Cancellation Rate	40.97%
Total Booking Value	\$12,101,899
Total Revenue at Risk	\$4,275,959
Revenue at Risk, action list	\$3,646,308

Optimization

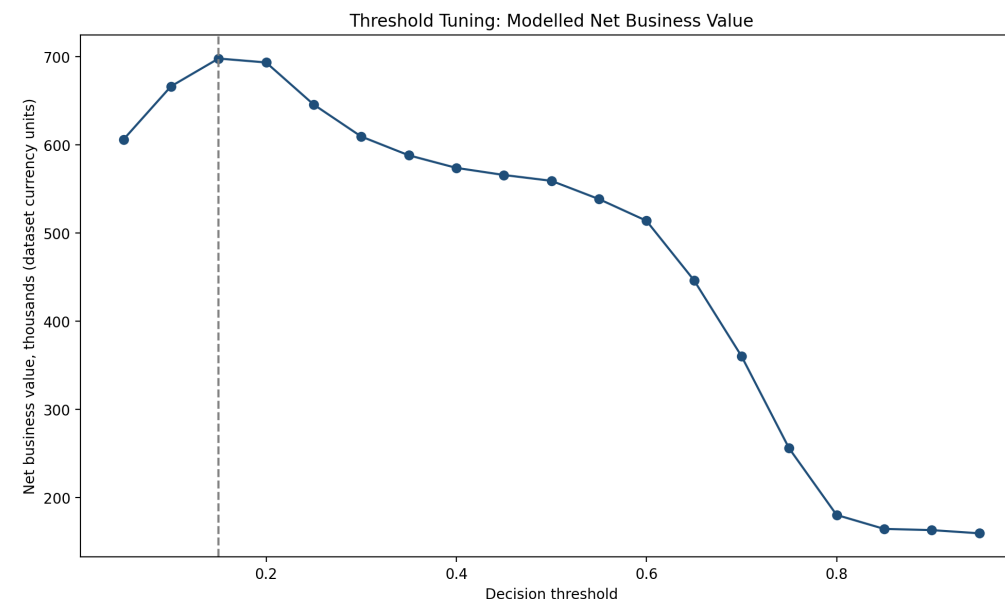
Five passes, and the first one did most of the work

Pass	What changed	Effect
1	Removed a calendar feature that could not generalise	ROC-AUC 0.8238 to 0.8798
2	Fixed the convergence failure in the baseline	114 s to 4 s, now converges
3	Tuned the decision threshold to 0.25	Recall 0.617 to 0.823
4	Rebased the risk bands on the new threshold	Bands align with the model
5	Rebuilt the reporting layer on a single table	All nine measures reconcile

Why the calendar feature had to go

- Training covers 22 months, the test period covers 5, and 4 of those were never seen
- The model had learned month specific behaviour that did not exist in the future

Ten rounds of hyperparameter search achieved almost nothing by comparison



Moving the threshold from 0.50 to 0.25 raises modelled net value from \$558,993 to \$645,509 on the test period, with no retraining.

Where We Landed, and What Remains

The recommendation to the business

0.8798

ROC-AUC on the held out period

82.3%

Of real cancellations caught

68.6%

Of flagged bookings really cancel

\$4.28 M

Revenue at Risk measured

Recommendation

Adopt the ranked action list as the daily queue for the revenue desk, starting with the Very High band

Retrain on current data from the hotel's own system before the actions are trusted operationally

Two findings worth keeping

The biggest gain came from removing a feature, not from tuning

The worst fault was in the reporting layer and was invisible on screen

What remains

Publish the dashboard to the Power BI Service and schedule the refresh

Replace the file handoff with a scheduled export from the reservation system

Calibrate the middle probability band, where the model understates risk

Close the feedback loop, record which actions were taken and whether the booking survived

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